State Variable Theory

#Chairing #Explanation

Given our ***Euler-Lagrange Formulation***:

$$B(q)\ddot{q} + C(\dot{q}, q)\dot{q} + G(q) = \Omega\tau + \phi_{diss}(\dot{q}, q, t) + \langle\lambda(t), f(q)\rangle$$

We can propose our **state variables**:

$$\begin{cases} q_a = q \\ q_b = \dot{q} \end{cases}$$

Then, the *dynamics* of the proposed **state variables** are:

$$\dot{q}_a = \dot{q} = q_b$$

$$\dot{q}_b = \ddot{q} = B^{-1}(q_a)[\Omega\tau + \phi_{diss}(q_b, q_a, t) +$$

$$= f(q_a, q_b) + g(q_a)(\Omega\tau + \langle\lambda(t), f(q_a)\rangle)$$

Where:
$g(q_a) = B^{-1}(q_a)$
$f(q_a, q_b) = -B^{-1}(q_a)(C(q_b, q_a)q_b + G(q_a))$

Now, in order to control the system using this representation, we need to propose a ***linear controller*** that *mimics* this **system dynamics**. Here we analyze two cases.

Proposal 1. Unperturbed System

Similar to the previously proposed system, our dynamics are expressed as:

$$\begin{cases} \dot{q}_a = q_b \\ \dot{q}_b = f(q_a, q_b) + g(q_a)(\Omega\tau + \langle\lambda(t), f(q_a)\rangle) \end{cases}$$

Now, we need to propose a controller so we can control our dynamics disregarding the **system**, in other words, for a ***PD controller*** we need:

$$\dot{q}_b = f(q_a, q_b) + g(q_a)(\Omega\tau + \langle\lambda(t), f(q_a)\rangle) = -K_P q_a - K_D q_b$$

To do this, we may simply propose a ***PD controller*** for the **torque** τ that takes into account the dynamics.

$$\tau = \Omega^{-1}[g^{-1}(q_a)(-K_P q_a - K_D q_b - f(q_a, q_b))] - \langle\lambda(t), f(q_a)\rangle$$

$$\tau = (J^T)^{-1}[B^{-1}(q_a)(-K_P q_a - K_D q_b - f(q_a, q_b))]$$

When substituting into the dynamics we get:

$$\dot{q}_b = f(q_a, q_b) + g(q_a)(\cancel{\Omega(\Omega^{-1}} [g^{-1}(q_a)(-K_P q_a - K_D q_b - f(q_a, q_b))] \cancel{- \langle \lambda(t), f(q_a) \rangle + \langle \lambda(t), f(q_a)}$$

$$\dot{q}_b = f(q_a, q_b) + \cancel{g(q_a)[g^{-1}(q_a)} (-K_P q_a - K_D q_b - f(q_a, q_b))]$$

$$\dot{q}_b = \cancel{f(q_a, q_b) - f(q_a, q_b)} - K_P q_a - K_D q_b$$

$$\dot{q}_b = -K_P q_a - K_D q_b$$

We basically made a ***controller*** for the **system's acceleration** by proposing one for the **torque**. Resulting in our new dynamics:

$$\begin{cases} \dot{q}_a = q_b \\ \dot{q}_b = -K_P q_a - K_D q_b \end{cases}$$

In **vector form**:

$$\frac{d}{dt}\mathrm{q} = \frac{d}{dt}\begin{bmatrix} q_a \\ q_b \end{bmatrix} = \begin{bmatrix} q_b \\ -K_P q_a - K_D q_b \end{bmatrix}$$

$$= \begin{bmatrix} 0_{n \times n} & I_{n \times n} \\ -K_P & -K_D \end{bmatrix} \begin{bmatrix} q_a \\ q_b \end{bmatrix}$$

$$\frac{d}{dt}\mathrm{q} = A\mathrm{q}$$

From this, is obvious to note that q is a 12×1 vector, and both K_P and K_D are 6×6 matrices.

Proposal 2. Perturbed System

Consider the case where:

$$\begin{cases} \dot{q}_a = q_b \\ \dot{q}_b = f(q_a, q_b) + g(q_a)(\Omega\tau + \langle \lambda(t), f(q_a) \rangle) + \psi(q_a, q_b, t) \end{cases}$$

Where:
$\psi(q_a, q_b, .)$: Non-modelled sections of the robot.
$\psi(., ., t)$: Effect of all external perturbations.

Where the perturbations ψ are constrained to a set Ψ defined by:

$$\Psi = \{\psi : Q_a \times TQ_a \times \mathrm{R}^+ \to \mathrm{R}^n | \; ||\psi||^2 \leq \psi_0 + \psi_1 ||q_a||^2 + \psi_2 ||q_b||^2\}$$

Where:

$Q_a \subset \mathrm{R}^n$: Configuration space (positions of q_a)

TQ_a : Tangent bundle of Q_a, representing positions and velocities (q_a, q_b)

$\psi_0, \psi_1, \psi_2 \geq 0$: Bounds on perturbation magnitude where:

ψ_0 : Constant Disturbances (e.g. sensor noise)

ψ_1 : Uncnertainties that scale with position.

ψ_2 : Velocity-dependent disturbances

Here, TQ_a represents all posible states (q_a, q_b) where $q_b = \dot{q}_a$ are velocities. It essentially **formalizes** the **state-space** of the *system* as pairs of *positions* and *velocities*.

We follow the same procedure to propose a **control input** τ to cancel *nonlinear* terms and enforce *linear dynamics*:

$$\tau = \Omega^{-1}[g^{-1}(q_a)(-K_P q_a - K_D q_b - f(q_a, q_b))] - \langle \lambda(t), f(q_a) \rangle$$

So we end up with:

$$\dot{q}_b = -K_P q_a - K_D q_b + \psi(q_a, q_b, t)$$

$$\frac{d}{dt} q = \underbrace{\begin{bmatrix} 0_{n\times n} & I_{n\times n} \\ -K_P & -K_D \end{bmatrix}}_{A} \begin{bmatrix} q_a \\ q_b \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ I \end{bmatrix}}_{B} \psi$$

Where we said our perturbations ψ are bounded by:

$$||\psi||^2 \leq \psi_0 + \psi_1 ||q_a||^2 + \psi_2 ||q_b||^2$$